Problem 1: Let $T : \mathbb{R}^3 \longrightarrow \mathbb{R}^3$ be a linear operator which has an upper triangular matrix with respect to the basis $\beta = ((1,0,0), (1,1,1), (1,1,2))$. Find an orthonormal basis of $\mathbb{R}^3$ with respect to which T has an upper triangular.

Digression

Let V be a finite dimensional inner product space and $\beta = (v_1, \ldots, v_n)$ s.t given $T \in \mathcal{L}(V)$, $[T]_\beta^\beta$ is upper triangular.

$$\Rightarrow \quad Tv_k \in \operatorname{span}(v_1, \ldots, v_k)$$

Apply Gram-Schmidt Orthonormalization to β and obtain $\beta' = (w_1, \ldots, w_n)$.

Then span $(w_1, \ldots, w_k)$ = span $(v_1, \ldots, v_k)$ $\forall\ 1 \le k \le n$.

We are interested in Tw_k

$$w_k \in \text{span}\,(v_1, \ldots, v_k)$$

$$w_k = a_1 v_1 + \cdots + a_k v_k$$

$$Tw_k = a_1 Tv_1 + \cdots + a_k Tv_k.$$

$$Tv_j \in \text{span}\{(v_1, \ldots, v_j)\} \subseteq \text{span}\{(v_1, \ldots, v_k)\}$$

$$= \text{span}\{(w_1, \ldots, w_k)\}$$

$$\forall\ 1 \le j \le k$$

$$\Rightarrow \quad Tw_k \in \operatorname{span}((w_1, \ldots, w_k))$$

$$\Rightarrow \quad [T]_{\beta'}^{\beta'} \text{ is upper triangular}$$

Solution to Problem 1 : $T \in \mathcal{L}(\mathbb{R}^3)$ upper triangular

w.r.t $\beta = (\underbrace{(1,0,0)}_{v_1}, \underbrace{(1,1,1)}_{v_2}, \underbrace{(1,1,2)}_{v_3})$

$$w_1' = v_1$$

$$w_2' = (1,1,1) - \frac{\langle (1,1,1), (1,0,0)\rangle}{1} \cdot (1,0,0)$$

$$= (1,1,1) - (1,0,0) = (0,1,1)$$

$$w_3' = (1,1,2) - \langle (1,1,2), (1,0,0)\rangle (1,0,0)$$

$$- \frac{\langle (1,1,2), (0,1,1)\rangle}{2} (0,1,1)$$

$$= (1,1,2) - (1,0,0) - \frac{3}{2}(0,1,1)$$

$$= (0,1,2) - \left(0, \frac{3}{2}, \frac{3}{2}\right)$$

$$= \left(0, -\frac{1}{2}, \frac{1}{2}\right) = \frac{1}{2}(0,-1,1)$$

$$w_1 = (1,0,0)$$

$$w_2 = \frac{1}{\sqrt{2}}(0,1,1)$$

$$w_3 = \frac{1}{\sqrt{2}}(0,-1,1)$$

T will be upper triangular w.r.t the orthonormal basis

$\beta' = (w_1, w_2, w_3)$. $\blacksquare$

<u>Exercise:</u> Let V be an finite dimensional inner product space and let β' be obtained by the Gram schmidt orthonormalization of a basis β.

The what can we say about $[I]_{\beta}^{\beta'}$

Problem 2: Let $W = \text{span}\{(1,0,i), (1,2,1)\}$ in $\mathbb{C}^3$. Then compute an orthonormal basis of $W^{\perp}$.

Solution: $\beta = \{\overset{v_1}{(1,0,i)}, \overset{v_2}{(1,2,1)}, \overset{v_3}{(0,0,1)}\}$ be a basis obtained by extending the basis of W.

Let $\beta' = (w_1, w_2, w_3)$ be obtained by the G-S orthonormalization. Then

$\{w_3\}$ will be a basis of $W^{\perp}$

(Notice that $\dim(W^{\perp}) = \dim(\mathbb{C}^3) - \dim(W)$

$= 3-2 = 1$)

$$\beta = \big((1,0,i),\ (1,2,1),\ (0,0,1) \big)$$

$$w_1' = (1,0,i)$$

$$w_2' = (1,2,1) - \frac{\langle (1,2,1), (1,0,i) \rangle}{2} (1,0,i)$$

$$= (1,2,1) - \frac{1-i}{2} (1,0,i)$$

$$= \left(\frac{1+i}{2},\ 2,\ \frac{1-i}{2}\right) = \frac{1}{2}\left(1+i,\ 4,\ 1-i\right)$$

$$\frac{1}{2} + 4 + \frac{1}{2} = 5$$

$$w_3' = (0,0,1) - \frac{\langle (0,0,1),\ (1,0,i)\rangle}{2}(1,0,i)$$

$$- \frac{\langle (0,0,1),\ (1+i,\ 4,\ 1-i)\rangle}{5 \times 2}\left(\frac{1+i}{2},\ 2,\ \frac{1-i}{2}\right)$$

$$= (0,0,1) + \frac{i}{2}(1,0,i) - \frac{i+1}{10}\left(\frac{1+i}{2},\ 2,\ \frac{1-i}{2}\right)$$

$$= (i/2,\ 0,\ 1/2) \quad - \quad ''$$

$$= \frac{i}{2}(1, 0, -i) - \left(\frac{i}{10}, \frac{i+1}{5}, \frac{1}{10}\right)$$

$$= \left(\frac{2i}{5}, -\frac{(i+1)}{5}, \frac{2}{5}\right)$$

$$\frac{\left(i, -\frac{(1+i)}{2}, 1\right)}{\left(1 + \frac{1}{2} + 1\right)^{1/2}} = \frac{\sqrt{2}}{\sqrt{5}}\left(i, -\frac{(1+i)}{2}, 1\right)$$

—— $\blacksquare$.

Problem 3: In $\mathbb{R}^3$, compute the distance of the vector $u = (2, 1, 3)$ from the subspace $W = \{(x, y, z) : x + 3y - 2z = 0\}$.

Solution: Let $\beta = \{(-3, 1, 0), (2, 0, 1)\}$ be a basis of W.

Let $\beta' = (w_1, w_2)$ be obtained by G-S-O.

$$w_2' = (2,0,1) - \frac{\langle (2,0,1),(-3,1,0)\rangle}{10}(-3,1,0)$$

$$= (2,0,1) + \frac{3}{5}(-3,1,0)$$

$$= \left(\frac{1}{5}, \frac{3}{5}, 1\right)$$

Then the projection of $(2,1,3)$

Let $v = \langle (2,1,3), w_1 \rangle w_1 + \langle (2,1,3), w_2 \rangle w_2$

$$\frac{1}{10} \langle (2,1,3), (-3,1,0) \rangle (-3,1,0) +$$

$$\frac{\langle (2,1,3), (1,3,5) \rangle}{35} (1,3,5)$$

$$= -\frac{1}{2}(-3,1,0) + \frac{4}{7}(1,3,5)$$

$$= \left(\frac{3}{2} + \frac{4}{7},\ \frac{12}{7} - \frac{1}{2},\ \frac{20}{7} \right)$$

$$= \left(\frac{29}{14},\ \frac{17}{14},\ \frac{20}{7} \right)$$

Hence distance of $(2,1,3)$ to W is given by

$$\left\| (2,1,3) - \left(\frac{29}{14},\ \frac{17}{14},\ \frac{20}{7} \right) \right\|$$

$$= \left\| \left(-\frac{1}{14}, -\frac{3}{14}, \frac{1}{7} \right) \right\|$$

$$= \left(\frac{1}{14^2} (1 + 9 + 4) \right)^{1/2}$$

$$= \frac{1}{\sqrt{14}}$$

Problem 4: Assume that $(v_1, \dots, v_k)$ is an orthonormal set in an inner product space V. Let $u \in U$. Then prove that

$$\sum_{j=1}^{k} |\langle u, v_j \rangle|^2 \leq \|u\|^2$$

with equality iff $u \in \operatorname{span}(v_1, \dots, v_k)$.

Proof: Let $v = \langle u, v_1 \rangle v_1 + \cdots + \langle u, v_k \rangle v_k$.

Let $w = u - v$

$\langle w, v_j \rangle = \langle u - v, v_j \rangle = \langle u, v_j \rangle - \langle u, v_j \rangle = 0$

$\Rightarrow \quad u - v \perp v_j \quad \forall \; j = 1, \ldots, k.$

If $W = \text{span}(v_1, \ldots, v_k)$

then $(u - v) \perp W$.

In particular $w \perp v$

Hence $u = v + w$

where $v \perp w$

$$\|u\|^2 = \|v\|^2 + \|w\|^2$$

$$\Rightarrow \quad \|u\|^2 \geqslant \|v\|^2$$

Recall $v = \langle u, v_1 \rangle v_1 + \cdots + \langle u, v_k \rangle v_k$

$$\|v\|^2 = \sum_{j=1}^{k} |\langle u, v_j \rangle|^2$$

with equality iff $w = 0$

$\iff \quad u = v$

$\iff \quad u \in W$. $\blacksquare$